

ONLINE STORE SALES DATA ANALYTICS PROJECT

Final Project Report

1. Project Overview

This project analyzes an online store sales dataset to understand product demand, city-level revenue, category performance, and monthly sales trends. The workflow covers data loading, data cleaning, revenue calculation, analysis, visualization, business insights, and recommendations.

Project Objective

- Identify which products sell the most.
- Identify the city generating the highest revenue.
- Compare revenue performance across product categories.
- Study monthly sales performance and trends.
- Identify areas where sales can be improved.

2. Dataset Description

The supplied source contains 2,000 order records with the following business fields: Order_ID, Date, Product, Category, City, Quantity, and Price. A Revenue field was created during processing using $\text{Quantity} \times \text{Price}$.

Field	Description
Order_ID	Unique order identifier
Date	Order date
Product	Product purchased
Category	Product category
City	Customer/order city
Quantity	Number of units sold
Price	Unit price
Revenue	Calculated as $\text{Quantity} \times \text{Price}$

3. Data Loading and Cleaning

The dataset was prepared for analysis using Pandas. The Date field was converted to datetime format, duplicate records were checked, missing values were reviewed, and data-quality anomalies were identified before calculating business metrics.

Data Quality Check	Result
Source records	2,000
Duplicate records	0
Missing Quantity	1
Non-positive Price records	42
Records used for final business analysis	1957

Data-quality note: the supplied source includes records with negative prices and one record (Order_ID 1005) where Quantity is blank. These records were excluded from the final business metrics because they do not represent valid positive sales quantities and prices. No duplicate records were found.

4. Data Processing

A new Revenue column was created using the formula:

Revenue = Quantity × Price

After data-quality filtering, 1,957 valid records were used for the main analysis. The valid records represent 5,948 units and ₹51,504,050 in calculated revenue.

5. Data Analysis Results

Total Revenue

The total calculated revenue from valid sales records is ₹51,504,050.

Best-Selling Product by Quantity

Chair is the best-selling product by quantity, with 568 units sold.

Best-Performing City by Revenue

Ahmedabad generated the highest revenue at ₹7,450,271.

Category Revenue Analysis

Category	Revenue	Share of Revenue
Electronics	₹37,093,350	72.0%
Furniture	₹13,343,784	25.9%
Accessories	₹905,769	1.8%
Stationery	₹161,147	0.3%

Electronics is the largest revenue-generating category, contributing 72.0% of total valid revenue.

Monthly Sales Analysis

Month	Revenue
2025-01	₹4,022,571
2025-02	₹3,655,451
2025-03	₹4,608,065
2025-04	₹4,783,813
2025-05	₹4,884,091
2025-06	₹3,654,222
2025-07	₹3,979,801
2025-08	₹4,012,636
2025-09	₹4,318,631
2025-10	₹4,429,234
2025-11	₹4,186,194
2025-12	₹4,891,509
2026-01	₹77,832

The highest monthly revenue in the cleaned dataset occurred in 2025-12, with ₹4,891,509.

6. Data Visualization

The following charts communicate the main business patterns identified in the analysis.

Revenue by City

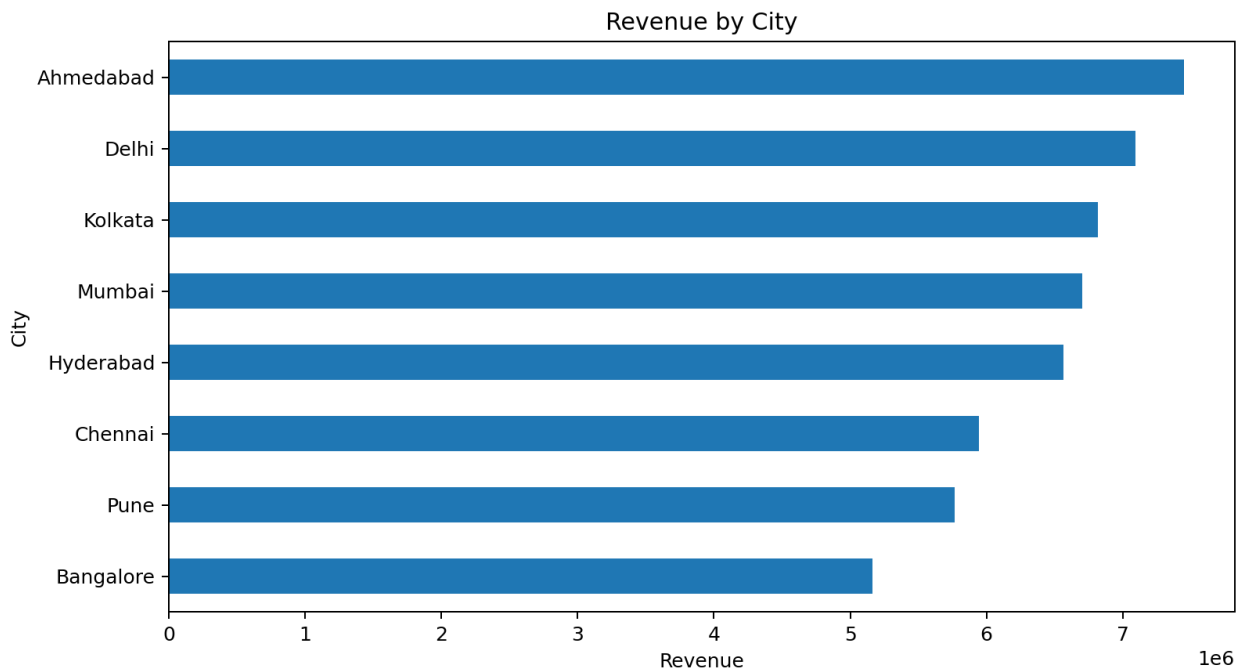


Figure 1. Revenue generated by city

Category Contribution

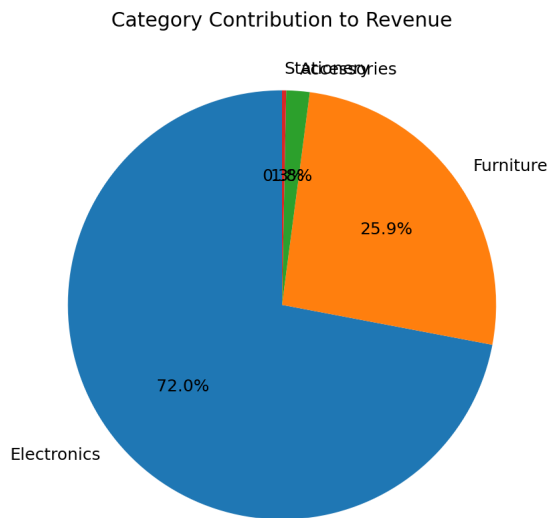


Figure 2. Category contribution to total revenue

Monthly Sales Trend

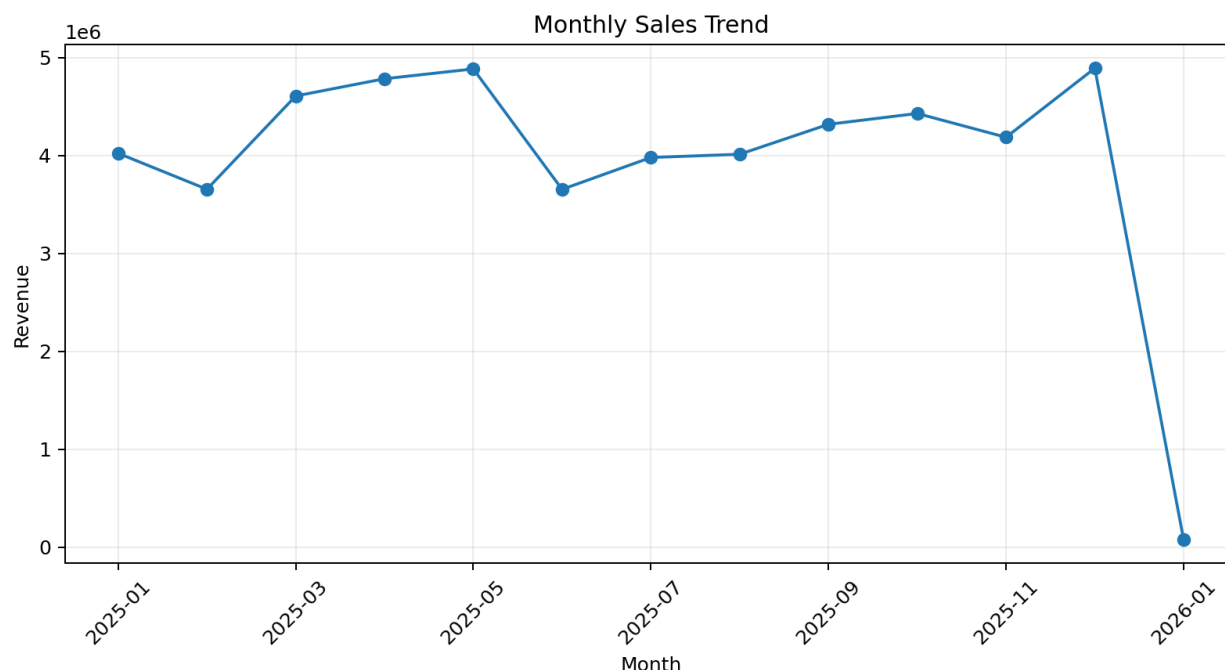


Figure 3. Monthly sales revenue trend

7. Business Insights

1. Chair has the highest unit sales, with 568 units, indicating strong demand by quantity.
2. Ahmedabad is the highest-revenue city, generating ₹7,450,271.
3. Electronics contributes 72.0% of valid revenue, making it the dominant revenue category.
4. Revenue is concentrated in higher-value products: the top revenue product is Laptop with ₹27,105,144.
5. 2025-12 is the strongest month in the analyzed period with ₹4,891,509 in revenue.
6. Bangalore has the lowest city revenue among the eight cities at ₹5,162,588, indicating an area where localized sales activity can be reviewed.

8. Business Recommendations

- Maintain stock availability and targeted promotions for Chair, while monitoring whether demand continues at the same level.
- Strengthen customer acquisition and marketing campaigns in lower-revenue cities, especially Bangalore, using localized offers and digital campaigns.
- Use the strongest months, particularly 2025-12, for planned promotions and inventory preparation, while testing offers during weaker months.
- Protect and grow the dominant Electronics category through product availability, cross-selling, and category-focused campaigns.

9. Project Deliverables

- Python Code (.py or Jupyter Notebook)

- Source sales dataset
- Charts for city revenue, category contribution, and monthly sales trend
- Final project report in Word/PDF format
- Optional extension: Power BI dashboard and SQL-based analysis

10. Conclusion

The analysis demonstrates a complete beginner-to-intermediate data analytics workflow: loading data, checking data quality, cleaning invalid records, creating a calculated Revenue metric, performing product/city/category/month analysis, and presenting the results through visualizations. The findings can be used to guide inventory planning, city-level marketing, category promotions, and seasonal sales campaigns.